

Karnataka's Mathematics Performance | Rural Government Schools

Submitted by: Team datathon_csf | Datathon 2026 | 1 August 2026

Executive summary¹

Every year, Akshara Foundation runs a maths contest in Gram Panchayats across Karnataka. Children in classes 4, 5 and 6 answer 20 questions. After evaluating the three years of data: 2022-23, 2023-24 and 2024-25, i.e. 13,79,087 answer sheets, here are the key insights and recommendations.

All of these children are in rural government schools. No private schools, aided schools or urban schools.

The Topline findings are :

- **38% of the eligible enrolment cohort sat for the assessment.** This is far from the census, but participation doubled across the three years (25.1% → 49.9%), so the assessment reach is expanding to cover all the eligible students eventually.
- **Children (Class 6) got much worse at multiplication and division.** Multiplication fell from 51% correct to 35%. Division fell from 55% to 38%. That is a big drop in three years.
- **The older children now do worse than the younger ones.** That is backwards. Normally a class 6 child should beat a class 4 child for the same competency. In both 2022-23 and 2023-24 they did, but by 2024-25 they did not.
- **Boys and girls move together.** The gap is small (a few percentage points), consistently in the same direction across all three years, and the standardised effect size (Cohen’s d) is negligible. Gender is not where the policy should focus on.
- **The vast majority of score variation is within Clusters and Districts, not GPs or Blocks.** State-level targeting, the default unit for most state interventions, cannot reach where it matters the most, and has a limited impact. Most of what separates a strong performer from a weak one based on this assessment happens inside a Cluster or a District.
- **Two independent assessments corroborate our district-level ranking.** Positive rank correlation with both ASER and PARAKH district scores indicates the instrument is measuring something real and externally consistent.
- **Kalyana Karnataka underperforms the rest of the state by a double-digit percentage-point margin,** at a small-to-medium standardised effect size. This is consistent with the region’s documented historical development gap under Article 371J, and is likely driven more by consistent lack of inputs that the encompassing districts have received with respect to the others.

Explanation Of Findings

1. 38% of the eligible enrolment cohort sat for the assessment

Out of all the children who could have taken this test, how many did? We counted every child enrolled in classes 4 to 6 in a rural government school, using the government's own UDISE+ records. Then we compared the assessment data.

Year	Grade 4	Grade 5	Grade 6	All grades
2022-23	25.4%	26.3%	23.6%	25.1%
2023-24	39.5%	40.7%	36.9%	39.0%
2024-25	49.1%	52.8%	48.0%	49.9%

¹ To reproduce every graph and raw data offline, execute `python src/run_all.py`. Every numeric claim is traceable through `claims.json`. The universe is rural State Government schools only, grades 4 to 6, 2022-23 through 2024-25 and the 13,79,087 assessment records supplied in the primary dataset. Every UDISE denominator filters to `management_group = State Government, Location = Rural, Grades = 4 to 6`

In the first year, only 1 child in 4 took the test. By the third year, 1 in 2. That is real progress and the teams who did it deserve credit.

Across the three years, total 13,79,087 assessments happened out of 36,44,154 eligible as per the enrolment. That is 38%.

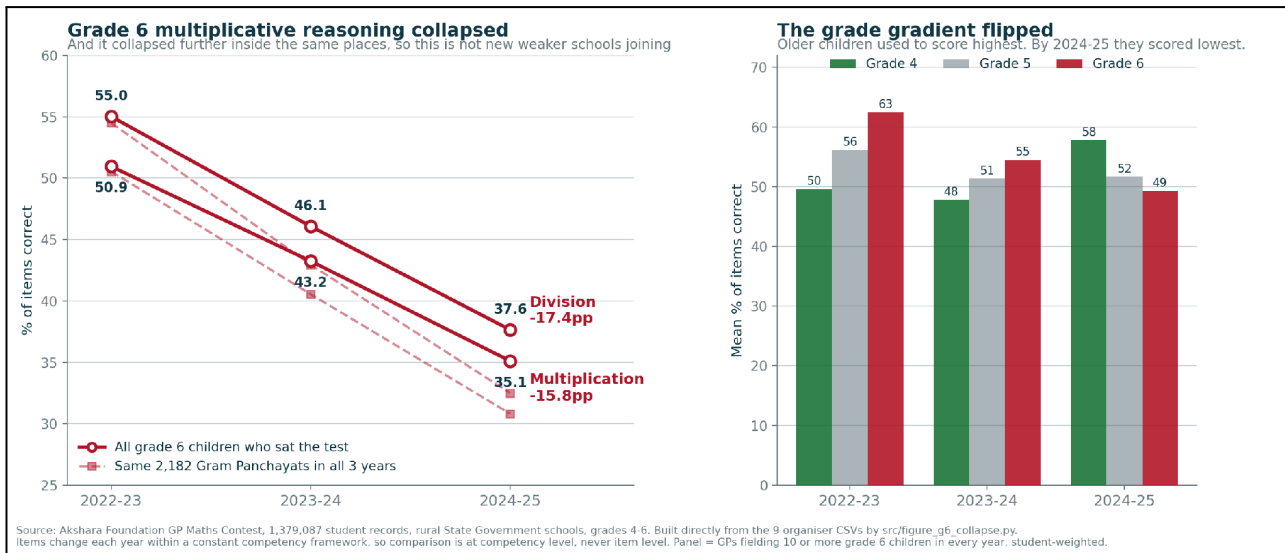
Which districts tested the most children, and which tested the least (three years pooled)

Top Rank	Best covered	Coverage	Bottom Rank	Least covered	Coverage
1	Chitradurga	70.1%	27	Udupi	5.7%
2	Dharwad	66.1%	28	Dakshina Kannada	2.0%
3	Bengaluru Rural	62.6%	29	Uttara Kannada	0.8%
4	Gadag	60.8%	30	Bengaluru Urban	0.0%
5	Chikkaballapura	59.1%	31	Shivamogga	0.0%

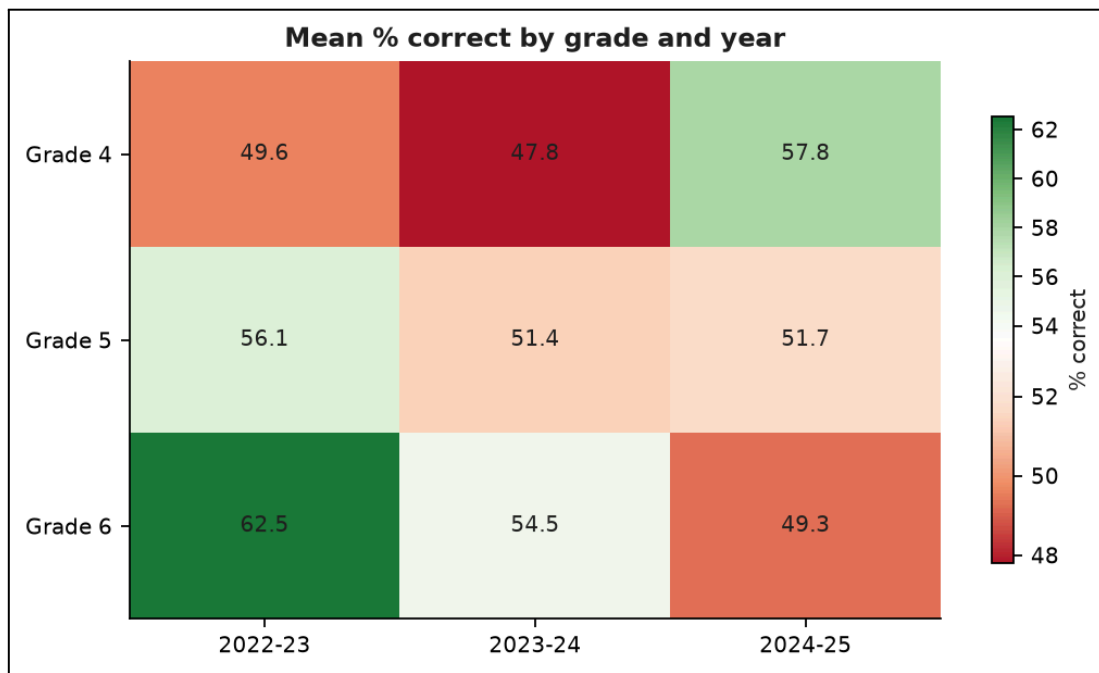
Two districts never took part at all, despite having some rural enrolment. Three more are effectively absent: Uttara Kannada's entire result rests on 801 children out of 105,160 enrolled. All three coastal districts skipped the 2024-25 round completely.

This immediately breaks the league table. Chitradurga, Dharwad, and Bengaluru Rural are the three highest covered districts while Udupi, Dakshina Kannada, Uttara Kannada are the three worst-covered districts in the state. Coverage and score correlate at $r = -0.47$ ($p = 0.009$, 29 districts): which means the better a district measures itself, the lower it performs.

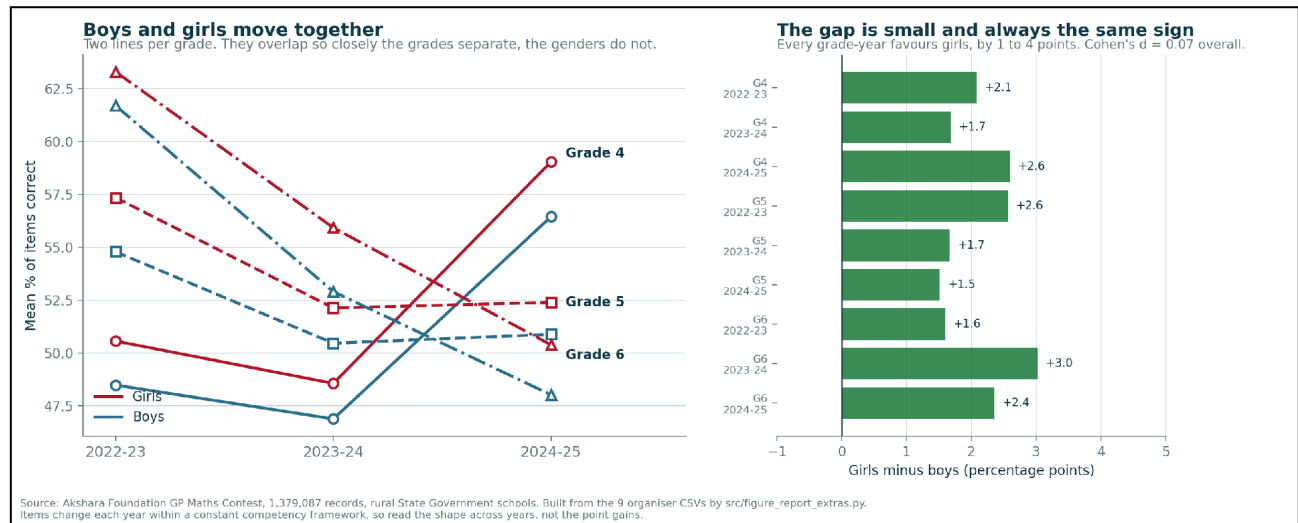
2. Children (Grade 6) got much worse at multiplication and division and the older children now do worse than the younger ones



In 2022-23 older children scored higher, as you would expect: grade 4 at 49.6%, grade 5 at 56.2%, grade 6 at 62.5%. By 2024-25 the order had reversed: grade 4 at 57.8%, grade 5 at 51.7%, grade 6 at 49.3%. The damage concentrated in multiplicative reasoning, where grade 6 multiplication fell from 51.0% to 35.1% and division from 55.0% to 37.6%.



3. Boys and girls move together



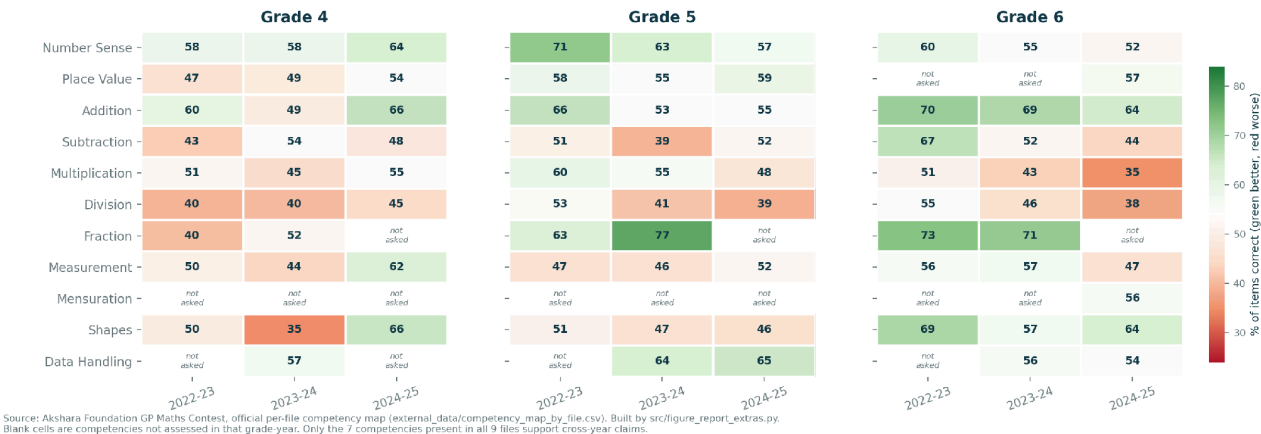
Split the same lines by gender and nothing separates. Girls lead in every one of the nine grade-years, but by 2.1 percentage points on average and never by more than 3.1. The grades pull apart; the genders do not. Overall, girls score 54.0% against boys' 52.0%, Cohen's $d = 0.071$, which means a randomly chosen girl beats a randomly chosen boy about 52 times in 100 instead of 50.

Girls are assessed at higher rates than boys (39.7% against 35.9%, and better covered in 26 of 31 districts), so we checked whether the lead is a participation artefact. It is not, and the check cuts in our favour. Where girls are over-assessed their measured lead is *smaller*, not larger ($r = -0.47$ across 80 district-years), because reaching deeper into girls' enrolment pulls in more marginal girls. In the 25 district-years where boys' and girls' coverage is within 2pp, the lead rises to +3.6pp. Differential participation deflates the reported gap rather than inflating it.

The practical reading: gender is real but small, and it is not where an intervention should start. The grade gradient is where the problem is.

4. All 11 skills across all three grades across all three years

Every competency, every grade, every year
Grade 6 darkens from left to right. Grade 4 lightens. Fraction disappears after 2023-24; mensuration appears only once.



Look at the class 6 panel on the right. It gets redder as you go left to right, year by year. Now look at class 4 on the left. It gets greener. The two are moving in opposite directions.

The 20 questions change every year, so we cannot compare question to question. We can compare competencies. Seven skills appear in all nine question papers, and only those seven are safe to compare across years: number sense, addition, subtraction, multiplication, division, measurement and shapes.

Across all three years, division is the weakest of the seven safe skills at 43% correct. Multiplication is next at 49%. Compare that with addition at 61%.

Skills stack on top of each other. Children who can do addition get subtraction right 72% of the time. Children who cannot do addition get subtraction right only 24% of the time. That is a 48 point difference. The same thing happens between multiplication and division, a 45 point difference.

So the order of teaching matters. You cannot fix division by drilling division. You have to fix what comes before it.

The hypotheses that decide the story

We tested 36 hypotheses. 19 held up, 3 were weak, 14 were discarded. The full register with methodology, effect, verdict and assumptions per hypothesis is in outputs/HYPOTHESIS_REGISTER.xlsx and outputs/tables/hypothesis_menu.csv.

5.1 The critical test: does the decline survive the participation story?

ID	Question	Result	Verdict
EH1	Does the fall survive inside GPs present all three years?	2,182 GPs: multiplication -19.7pp, division -22.0pp, worse than the raw -15.8 and -17.4	Held up
EH3	Does it appear where coverage barely moved?	8 stable districts: -18.7pp multiplication, -21.4pp division	Held up
EH5	What remains after controlling coverage change?	-19.3pp predicted at zero coverage change (25 districts)	Held up
EH6	Does it appear in already deeply covered districts?	11 districts at 40%+ in 2022-23: -19.8pp	Held up
EH4	Do districts that expanded most lose most?	r = -0.18, p = 0.40. The selection story's own test fails	Discarded

EH22	Do newly reached GPs have weaker children?	New GPs score 53.1% against veterans' 48.6%, higher by 4.6pp	Discarded
EH2	Does the grade 4 improvement survive the same panel?	+5.6pp raw becomes -0.6pp. Compositional	Discarded

Six designs, each with different weaknesses, all landing between 18 and 22 points. Two of them point the opposite way from the selection story: expansion flattered the raw trend rather than causing the decline. And the same panel that confirms the bad news deletes our own good news, which is why we lead with EH2 rather than bury it.

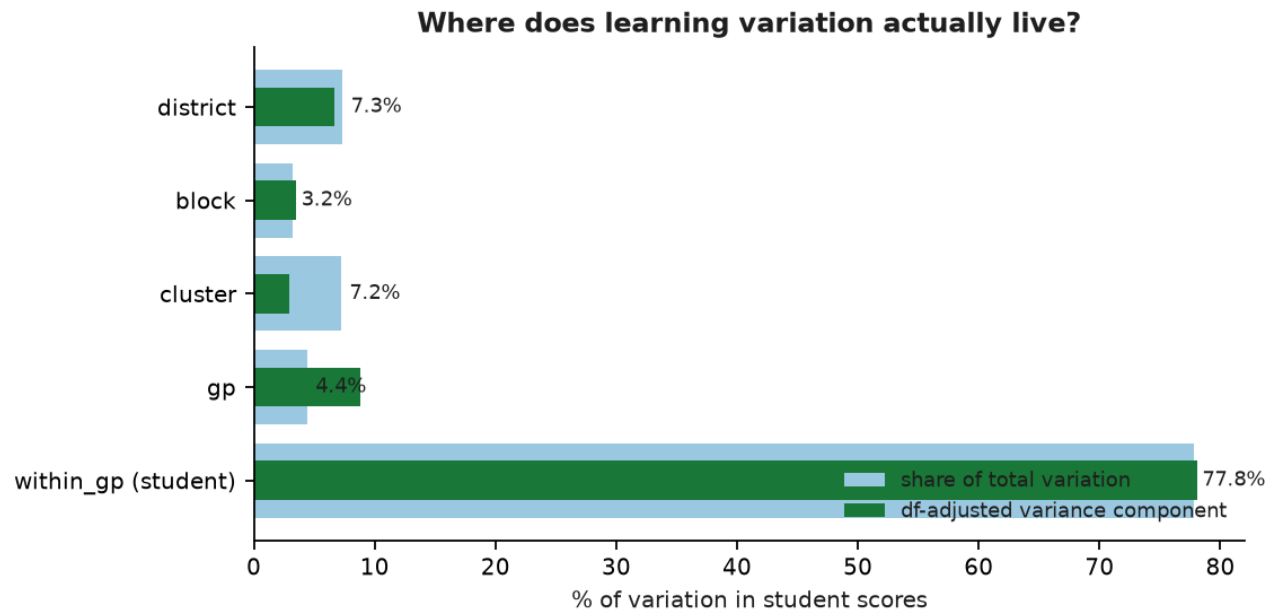
5.2 Other decisive tests

ID	Question	Result	Verdict
EH11	Does our district map match ASER 2024, rural to rural?	Spearman rho = 0.56 (p = 0.004), Std 6-8 division	Held up
EH9	Do girls lead where assessment is gender-balanced?	+3.6pp across 25 balanced district-years, above the +2.8 overall	Held up
H5	Is the Kalyana Karnataka gap explained by inputs?	Raw -13.3pp, falls to -4.8pp after PTR, literacy and stunting	Held up
EH19	Is that gap closing?	Widening: -10.0, -11.9, -15.8pp across the three years	Held up
EH7	Does adjusting for coverage change the league table?	17 of 26 districts move 3 or more ranks	Held up
EH14	Is there a prerequisite ladder?	Addition to subtraction +48pp lift; multiplication to division +45pp	Held up
H12	Does it match PARAKH RS 2024 government schools?	rho = 0.51 (p = 0.004)	Weak but corroborating
EH8	Does over-assessing girls inflate their lead?	r = -0.47. It shrinks the lead instead	Discarded
EH20	Are any test items bad?	0 of 180 fall below 0.15 discrimination. The instrument is clean	Discarded
H4	Does NFHS-5 stunting predict this exact cohort?	r = -0.04. Our best-designed question returned nothing	Discarded
H1	Does block PTR predict scores on its own?	r = -0.18, p = 0.07. Not enough alone	Discarded
H2	Do non-Kannada-medium belts score differently?	r = -0.01	Discarded
EH17	Did 2023 drought districts decline more?	The declaration covered 223 taluks. Untestable at district level	Discarded

6. Where the gaps are, and what sits behind them

6.1 The learning gap lives inside Gram Panchayats

78.1% of the variation in student scores lies within Gram Panchayats. The whole geographic hierarchy explains under 22%: district 6.7%, block 3.5%, cluster 2.9%, GP 8.8%.



The Targeting Efficiency Ceiling follows. An intervention perfectly targeted at district level can address at most about 6.7% of the variation; even perfect GP-level targeting reaches 21.9%. District spread is wide (32.3pp between top and bottom district means), but the median within-district GP spread is 28.2pp, nearly as large as the state's entire range.

4.2 Kalyana Karnataka: the gap has a reason

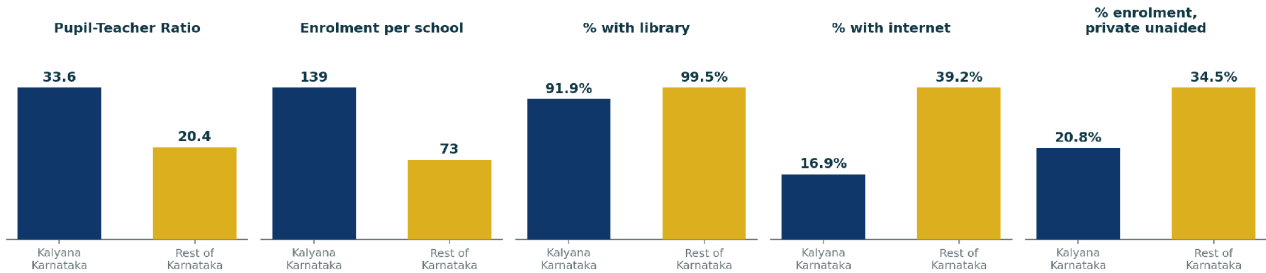
The 7 districts covered by Article 371J score 13.3 points below the rest of Karnataka. And the gap is getting worse: 10 points behind in 2022-23, 12 points in 2023-24, 16 points in 2024-25.

But when we account for teachers, adult literacy and childhood nutrition, the gap shrinks to 4.8 points. So most of the gap is not a mystery. It is resources.

Here is what those resources actually look like.

Kalyana Karnataka against the rest of Karnataka: the UDISE+ input reality, 2024-25

More children per teacher, nearly twice the school size, less than half the internet. The score gap has a supply side.



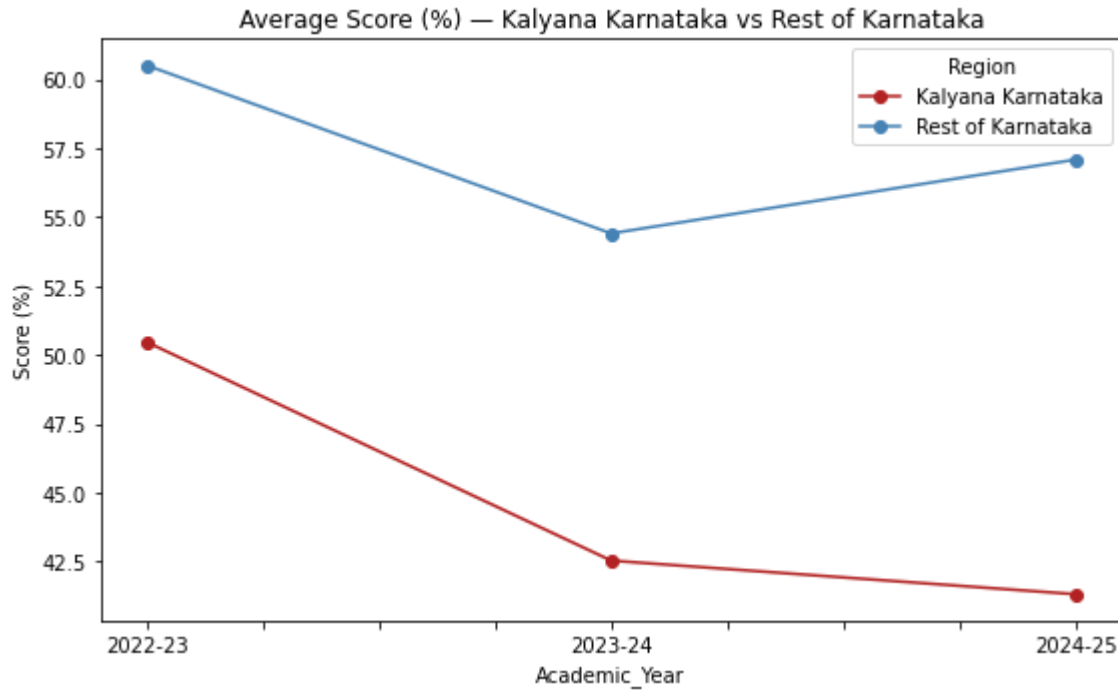
Source: UDISE+ 2024-25 school records (data/udise_csv), 371J flag from external_data/karnataka_district_crosswalk.csv. Built by src/figure_kk_vs_rest.py. First four metrics: rural State Government schools, the contest's universe. Private-unaided share: rural schools, all managements.

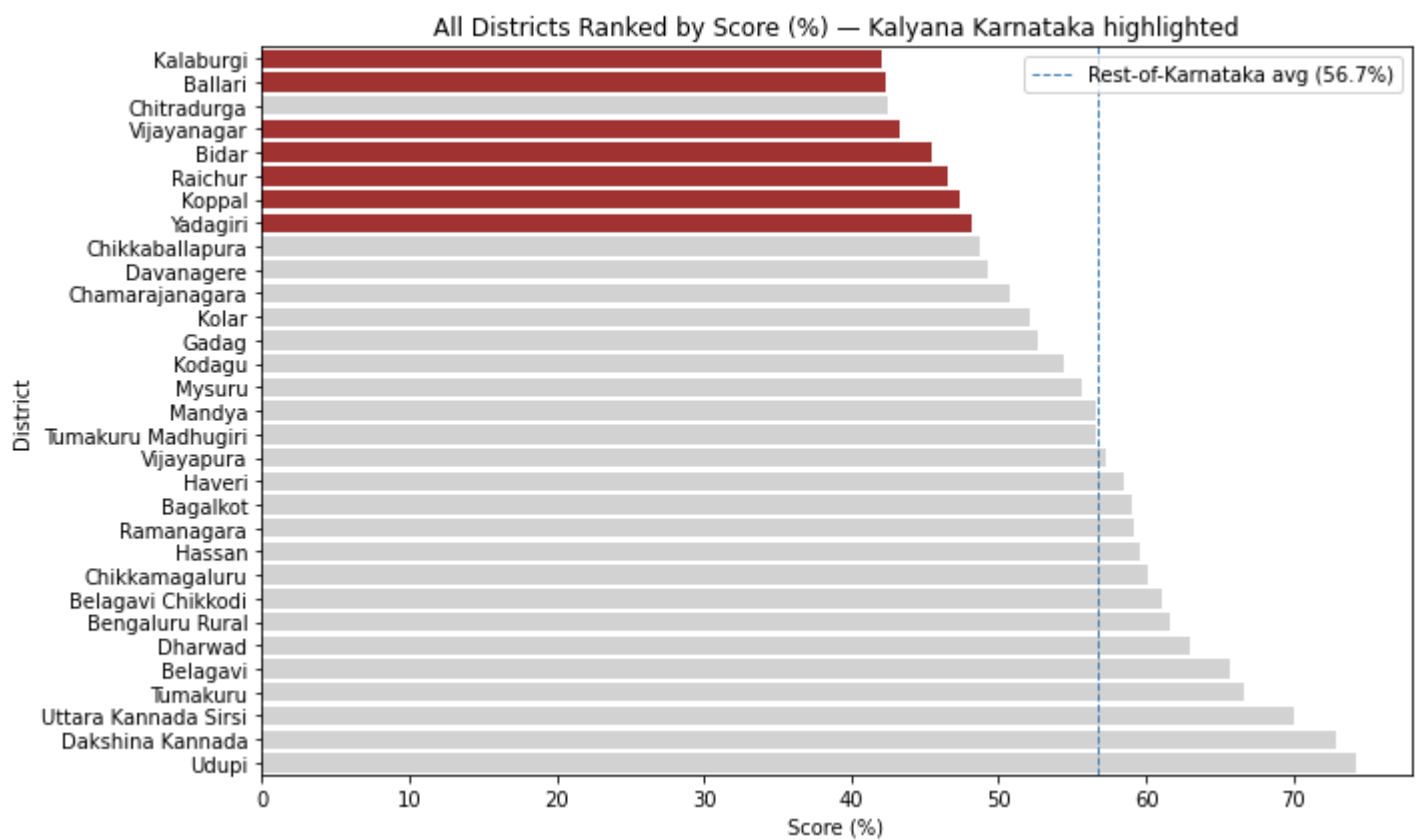
In rural government schools:

- **33.6 children per teacher**, against 20.4 in the rest of Karnataka. That is 13 more children for every teacher.
- **139 children per school**, against 73. Nearly twice as big.
- **Internet in 16.9% of schools**, against 39.2%. Less than half.
- Libraries are almost equal, 91.9% against 99.5%.

One more number that cuts the other way: only 20.8% of rural children there go to private schools, against 34.5% elsewhere. So government schools in Kalyana Karnataka teach a **larger** share of all children, with far fewer teachers per child.

This is not about effort or culture. It is a resourcing pattern, and a government can change resourcing.





This is not a mystery of culture or effort. It is a resourcing pattern, and resourcing is something a government can change. That is what makes recommendation 1 actionable.

6.3 Best and worst performing geographies, across four scales

The same top/bottom exercise run at District, Block, Cluster, and GP level tells a coherent story at the top two scales and starts to break down at the bottom two, where sample size stops supporting a ranking at all. Both things are worth reporting, and the point at which the ranking stops being trustworthy is itself a finding.

District. All seven Article 371J districts — Kalaburgi, Ballari, Vijayanagar, Bidar, Raichur, Koppal, Yadagiri — occupy the bottom ten of thirty-one. That is not a sample of Kalyana Karnataka's weakest performers; it is the entire region, with nothing left over. This is the same 13.3-point gap from section 4.2, now visible as a complete sorting rather than an average. At the top, three of the four best-scoring districts — Udupi, Dakshina Kannada, Uttara Kannada Sirsi — are the same three districts flagged in section 1 as the worst-covered in the state (coverage $r = -0.47$ with score). Their position here should be read through that caveat, not around it.

Rank	Best-performing district	Mean	n		Rank	Worst-performing district	Mean	n
1	Udupi	74.3	2,815		1	Kalaburgi	42.0	91,823
2	Dakshina Kannada	72.9	1,623		2	Ballari	42.3	46,061
3	Uttara Kannada Sirsi	70.0	801		3	Chitradurga	42.4	87,338
4	Tumakuru	66.7	52,084		4	Vijayanagar	43.3	61,409
5	Belagavi	65.7	56,630		5	Bidar	45.4	42,933

6	Dharwad	62.9	66,812	6	Raichur	46.6	70,124
7	Bengaluru Rural	61.6	27,631	7	Koppal	47.4	63,307
8	Belagavi Chikkodi	61.1	59,062	8	Yadagiri	48.2	44,853
9	Chikkamagaluru	60.1	28,894	9	Chikkaballapura	48.7	36,469
10	Hassan	59.6	45,217	10	Davanagere	49.3	38,336

Every block on this table carries a large enough n to trust. Byndoor tops it at 85.1%, corroborating the bright-spot residual already reported in section 4.4. The bottom is dominated by Kalyana Karnataka blocks — Chittapur, Sedam, Gulbarga South, Chincholi (all Kalaburgi), Siruguppa (Ballari), Harapanahalli (Vijayanagar) — again the district-level gap resolving into its constituent parts. Molakalmur, in non-KK Chitradurga, is the single lowest block in the state at 29.9%.

Rank	Best-performing block	Mean	n	Rank	Worst-performing block	Mean	n
1	Byndoor	85.1	560	1	Molakalmur	29.9	10,764
2	Yellapur	83.4	335	2	Chittapur	35.2	13,761
3	Devanahalli	81.2	5,936	3	Challakere	37.5	21,467
4	Ramanagara	76.3	4,460	4	Sedam	37.9	10,516
5	Talikoti	76.0	298	5	Jagalur	38.6	7,288
6	Chikodi	74.6	7,423	6	Siruguppa	39.5	14,014
7	Brahmavara	73.5	1,850	7	Ballari West	40.3	6,611
8	Mangaluru South	72.9	1,623	8	Gulbarga South	40.4	7,553
9	Gubbi	72.5	11,721	9	Harapanahalli	41.3	17,073
10	Pavagada	72.3	3,857	10	b	41.4	12,003

Cluster and GP — where the ranking stops being trustworthy without a floor. Both tables in the source data were pulled with no minimum-n filter, and it shows. The worst-performing cluster, Indi (Urdu), is a single child (n = 1, std undefined); Challakere North is two children; Urdu Maganalli is six. A "cluster average" built on one to six students is not a finding about that cluster, it is a data point about that data point. The same problem appears at the top: Urdu Naragund (n = 4), C Kunnala (n = 20), and Padukone (n = 9) cannot support a claim that they are genuinely among the state's best. None of these belong in a published ranking without a reliability floor applied first — the same discipline the report already applies to external covariate joins in section 6 (student-weighted match $\geq 60\%$ before a claim is allowed).

Filtering to $n \geq 100$ changes the bottom-cluster picture substantially: Nagasamudra (n = 1,401, mean 21.2%) and Hanagal (n = 1,025, mean 24.0%) are the only two clusters in the bottom ten large enough to trust outright, and Nagasamudra is worth flagging specifically — it recurs as a bottom performer at the GP level too (n = 639, mean 20.0%), and Bagdal shows the same cross-level persistence (cluster n = 303, mean 21.7%; GP n = 281, mean 19.0%). A locality that scores at the bottom at two different aggregation levels, each with a few hundred students behind it, is a genuine hyper-local crisis point rather than small-n noise — exactly the kind of GP-level target section 4.1's variance decomposition says district-level programmes cannot reach.

decomposition says district-level programmes cannot reach.

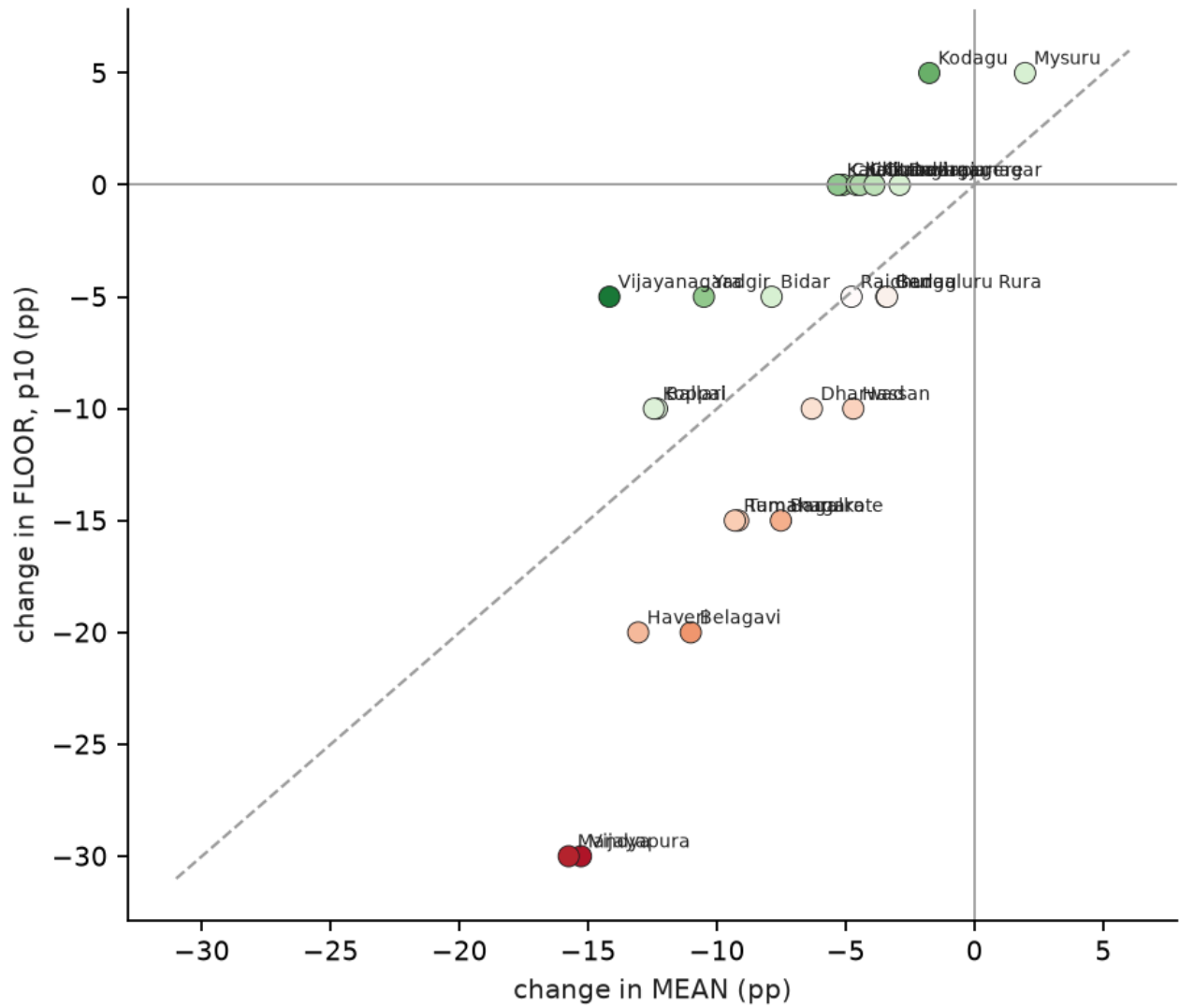
Best cluster	Mean	Worst cluster	Mean
Halaga	92.86	Sadam West	22.81
Navunda	90.95	Hanagal	24.0

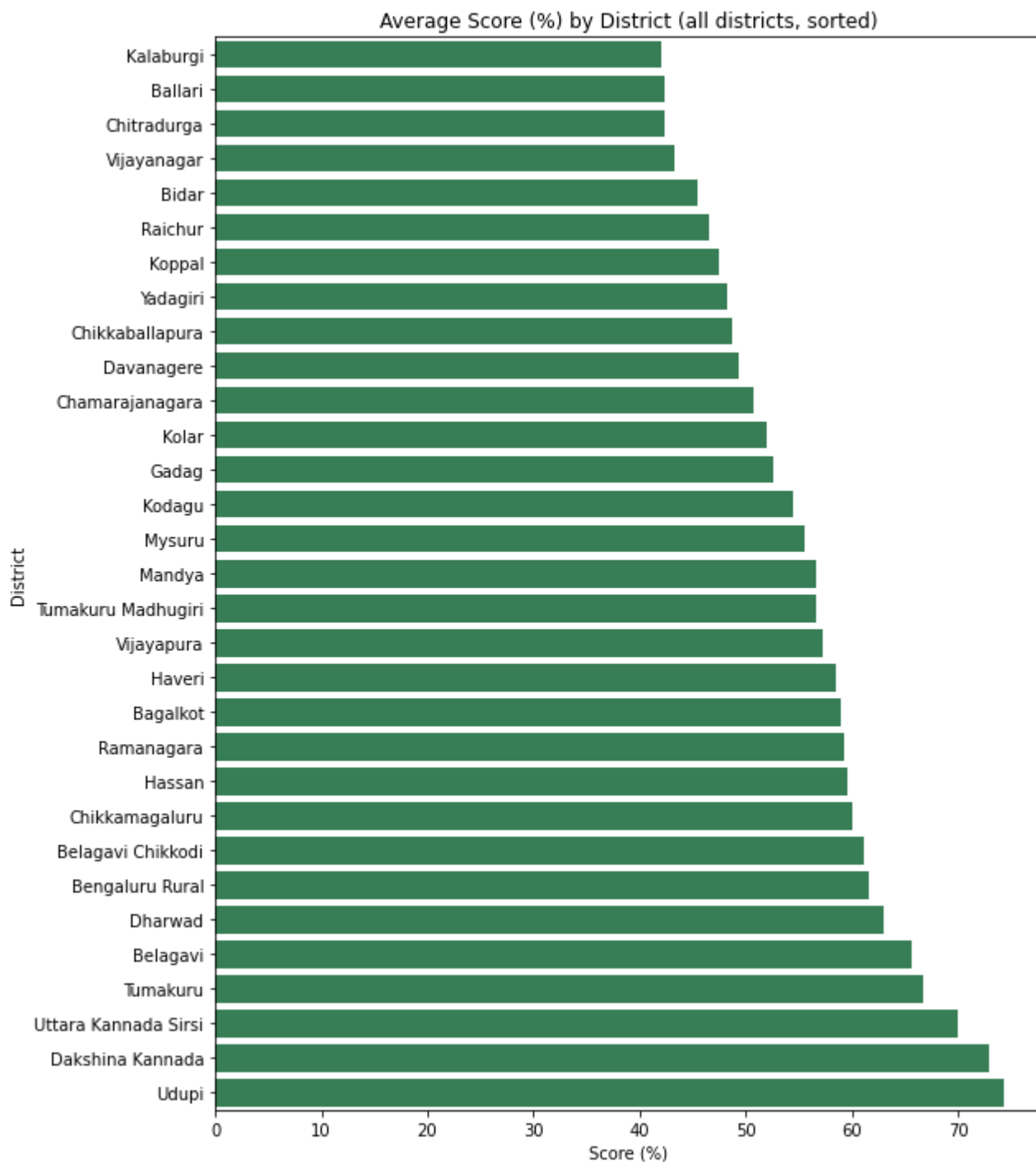
Best GP	Mean	Worst GP	Mean
Muniyuru	90.9	Bagdal	19.0
Kanchugaranahalli	90.2	Nagasamudra	20.0
Nandolli	90.3	Hongunta	20.0

6.4 Bright spots

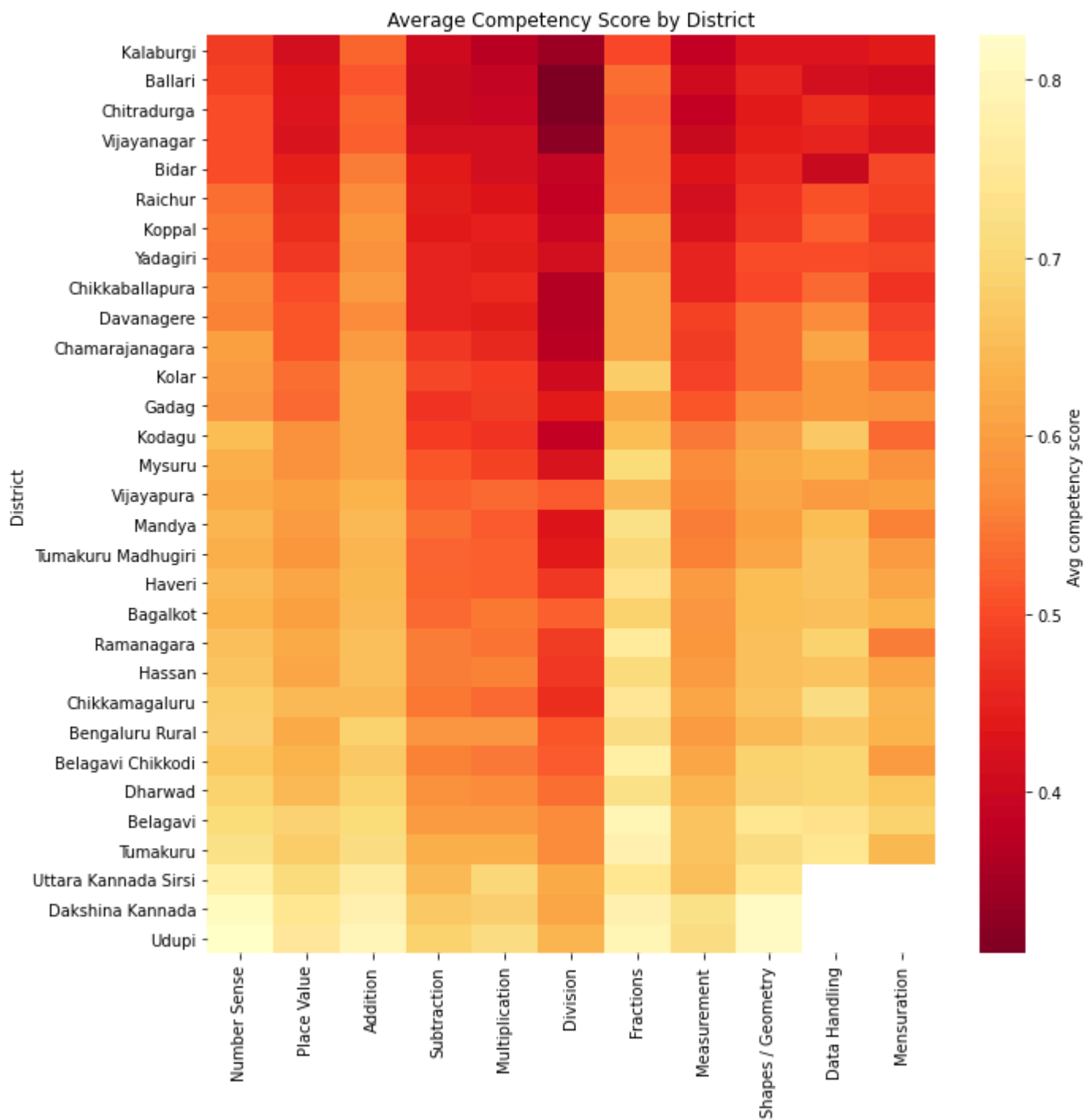
Byndoor block in Udupi scores 85.1% against 53.6% predicted from its structural conditions, a residual of +31.5pp. Kodagu's blocks underperform their conditions badly (Madikeri -20.9pp, Virajpet -20.3pp).

Did the average move, or did the weakest children move? Below the line = growth that skipped the bottom





Bright spots are not scattered at random: they cluster (chi-square $p = 0.022$), with Belagavi holding 6 of the 22 flagged blocks. The block prediction model explains $cv_R2 = 0.26$ of variation, so residuals are leads for a field visit, not verdicts.

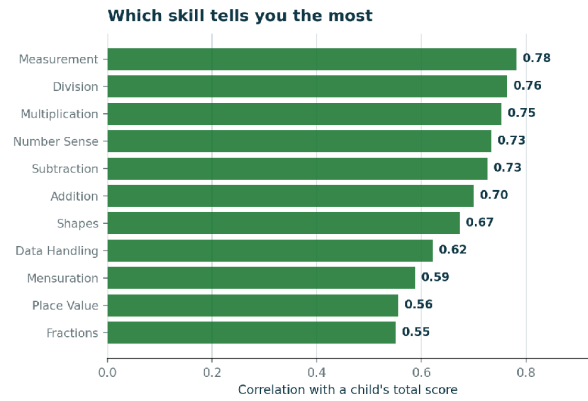
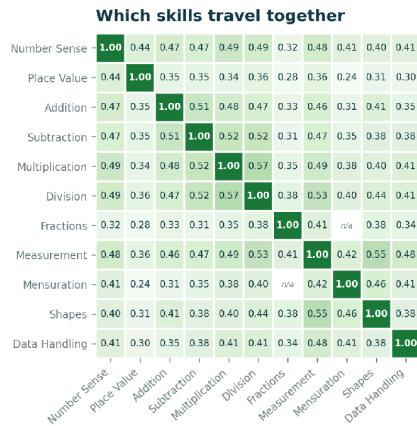


5. Which skills matter most

We looked at how the 11 skills relate to each other, and to a child's total score.

How the 11 skills relate to each other and to the whole test

Darker means the two skills move together more closely. The bars show which single skill best predicts a child's overall score.



Source: Akshara Foundation GP Maths Contest, 1,379,087 student rows pooled across all 9 files, official competency map. Built by src/figure_competency_correlation.py. The bar chart includes each skill inside the total, so those numbers run higher than a like-for-like comparison would. Fractions and Mensuration never appear for the same child, so that one pair is undefined.

Which single skill tells you the most about a child? Measurement (0.78), then division (0.76), then multiplication (0.75). At the other end, place value (0.56) and fractions (0.55) tell you the least.

Note: these numbers include the skill inside the total score, so they are a bit generous. But the ranking is what matters.

Which skills travel together? Multiplication and division are the closest pair (0.57). Then measurement and shapes (0.55). Then subtraction with multiplication and with division (both 0.52).

One skill behaves oddly. Place value barely relates to anything else. Its strongest link is with number sense, at just 0.44, and its weakest is 0.24. A child's place value score tells you almost nothing about the rest of their test. For a foundational skill, that is strange, and the assessment team may want to look at how those questions are worded.

One box in the grid is empty, not zero. Fractions and mensuration never appear on the same paper, so we cannot compare them. That is a gap in the data, not a finding.

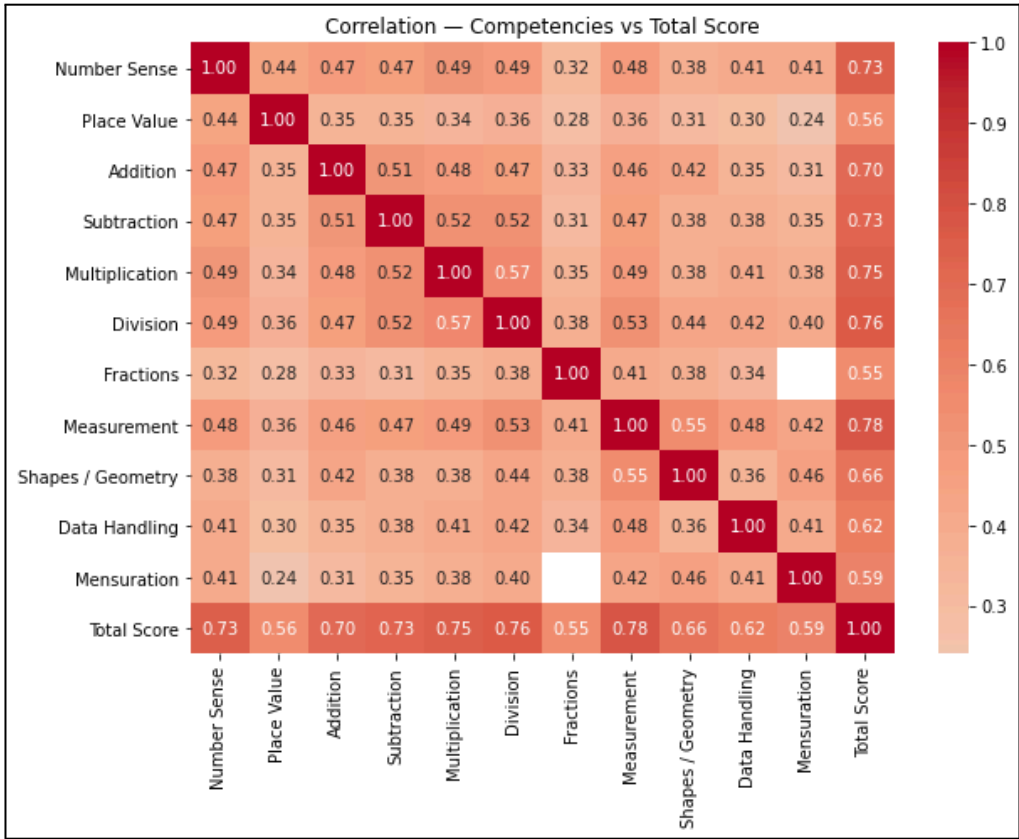
The useful conclusion: multiplication and division are not just where scores collapsed. They are also where the test carries the most information about a child. A catch-up programme built on them is aimed at the load-bearing part of maths, not a corner of it.

6.5 Triangulation

Two independent assessments agree with the contest's map. District means correlate with ASER 2024 rural Std 6-8 division at $\rho = 0.56$ ($p = 0.004$, 25 districts), rural against rural. District ranks replicate in PARAKH RS 2024 government-school grade-6 maths at $\rho = 0.51$.

PARAKH also supplies the arc our data lives inside: Karnataka government schools ahead of private at grade 3 (59 against 55), behind by grade 6 (40 against 47), with the state maths average falling 57 to 45 to 33 across grades 3, 6 and 9. Three instruments, one shape: mathematics is lost in upper primary, at the multiplicative step. Karnataka enters this window already behind the country, with 20.9% of rural Std V children able to divide in ASER 2024 against 30.7% nationally.

7. Inferential and Predictive Analysis



Correlation of competencies on total scores revealed that every competency correlates positively with Total Score, but not equally. Measurement carries the most information about the whole test ($r = 0.78$), followed by Division (0.76) and Multiplication (0.75); the multiplicative competencies and Measurement together anchor the score more than any other pair. Place Value (0.56) and Fractions (0.55) carry the least — roughly the weakest signal of any item in the framework.

Among competencies themselves, the strongest pairwise correlation is Multiplication–Division at 0.57, consistent with the prerequisite ladder already established in section 2.3, followed by Measurement–Shapes/Geometry at 0.55 and Subtraction–Multiplication and Subtraction–Division, both at 0.52. Place Value is the outlier: it correlates weakly with every other competency, from 0.24 with Mensuration to 0.44 with its nearest curricular neighbour, Number Sense. A child's Place Value score carries almost no information about how they perform elsewhere on the test, which is unusual for a foundational-numeracy item and worth a look from the assessment design team, not just the analysis team.

One cell in the matrix is blank rather than low: Fractions and Mensuration never co-occur for the same child, because Fractions is absent from every 2024-25 file and Mensuration exists only as a single grade-6 2024-25 question, both already noted in section 2.3. The correlation is undefined, not zero — a data-availability artefact, not a finding.

The practical reading: Division and Multiplication are not just where the score collapsed), they are also where the test's signal concentrates. A remediation programme built around the multiplicative ladder, as recommendation 2 proposes, is targeting the load-bearing part of the assessment, not a peripheral one. These were further explored using **simple linear regression, multivariate regressions, and random forest predictive models.**

Simple Linear Regression

	Competency	Competency_Label	Correlation	R_squared_univariate	n
7	MEA	Measurement	0.782	0.612	1379087
5	DIV	Division	0.764	0.583	1379087
4	MUL	Multiplication	0.753	0.568	1379087
0	NS	Number Sense	0.734	0.538	1379087
3	SUB	Subtraction	0.726	0.527	1379087
2	ADD	Addition	0.700	0.490	1379087
8	SHAPE	Shapes / Geometry	0.660	0.435	1379087
9	DH	Data Handling	0.622	0.387	856518
10	MENSU	Mensuration	0.588	0.346	181653
1	PV	Place Value	0.556	0.309	1131626
6	FRAC	Fractions	0.551	0.304	783559

Simple linear: what each competency explains alone

Every competency correlates positively with Total Score, but not equally. Measurement carries the most information about the whole test ($r = 0.78$), followed by Division (0.76) and Multiplication (0.75); the multiplicative competencies and Measurement together anchor the score more than any other pair. Place Value (0.56) and Fractions (0.55) carry the least — roughly the weakest signal of any item in the framework.

Among competencies themselves, the strongest pairwise correlation is Multiplication–Division at 0.57, consistent with the prerequisite ladder already established in section 2.3, followed by Measurement–Shapes/Geometry at 0.55 and Subtraction–Multiplication and Subtraction–Division, both at 0.52. Place Value is the outlier: it correlates weakly with every other competency, from 0.24 with Mensuration to 0.44 with its nearest curricular neighbour, Number Sense. A child's Place Value score carries almost no information about how they perform elsewhere on the test, which is unusual for a foundational-numeracy item and worth a look from the assessment design team, not just the analysis team.

One cell in the matrix is blank rather than low: Fractions and Mensuration never co-occur for the same child, because Fractions is absent from every 2024-25 file and Mensuration exists only as a single grade-6 2024-25 question, both already noted in section 2.3. The correlation is undefined, not zero — a data-availability artefact, not a finding.

The practical reading: Division and Multiplication are not just where the score collapsed (section 2.1), they are also where the test's signal concentrates. A remediation programme built around the multiplicative ladder, as recommendation 2 proposes, is targeting the load-bearing part of the assessment, not a peripheral one.

Multivariate Regression

Overall model $R^2 = 0.989$

	Competency	Competency_Label	Standardized_Coef
7	MEA	Measurement	1.308
4	MUL	Multiplication	1.116
0	NS	Number Sense	1.088
5	DIV	Division	0.975
3	SUB	Subtraction	0.936
2	ADD	Addition	0.803
8	SHAPE	Shapes / Geometry	0.801
1	PV	Place Value	0.424
9	DH	Data Handling	0.382
6	FRAC	Fractions	0.362
10	MENSU	Mensuration	0.093

Multivariate: what each competency explains once the others are held constant

A standardised linear regression of Total Score on all eleven competencies together reaches $R^2 = 0.989$ — essentially a perfect fit. That number is not a discovery about learning; it is a discovery about arithmetic. Total Score is a weighted sum of the competency subscores by construction (section 4.6 established that Total Score tracks the sum of the eleven competency scores almost exactly), so a linear model recovers that relationship rather than uncovering it. The standardized coefficients below describe how the scoring rubric weights each competency, not how each competency causes performance.

Read that way, Measurement carries the heaviest weight (1.308), ahead of Multiplication (1.116), Number Sense (1.088), and Division (0.975); Subtraction, Addition, and Shapes/Geometry cluster in the middle (0.936, 0.803, 0.801). Place Value, Data Handling, Fractions, and Mensuration all sit well below one (0.424, 0.382, 0.362, 0.093) — the same four competencies that were incomplete in section 4.7. Their missing values were imputed with the within-grade median before this model ran, which compresses their variance and understates their true weight in the rubric; their coefficients here are a floor, not a precise estimate.

Random Forest Predictive Modelling

Random Forest R^2 (train): 0.953

	Competency	Competency_Label	RF_Importance
5	DIV	Division	0.535
7	MEA	Measurement	0.212
0	NS	Number Sense	0.082
4	MUL	Multiplication	0.072
3	SUB	Subtraction	0.047
2	ADD	Addition	0.032
8	SHAPE	Shapes / Geometry	0.020
1	PV	Place Value	0.000
9	DH	Data Handling	0.000
6	FRAC	Fractions	0.000

Random Forest: a non-linear cross-check, and where it disagrees

A Random Forest trained on the same imputed feature set reaches $R^2 = 0.953$ on the training data — reported here as a diagnostic, not a validated model, since no held-out test split was used. Its importances tell a sharply different story from the standardized coefficients: Division alone accounts for 53.5% of total importance, Measurement for 21.2%, and the remaining nine competencies share the rest. Place Value, Data Handling, and Fractions register 0.000 importance — the model ignores them entirely.

This is a real disagreement, and it is explained by the same imputation gap flagged in section 4.8, not by those competencies being genuinely irrelevant. A Random Forest splits greedily on whichever feature reduces error most at each node; once Division and Measurement are available — both fully observed on all 1,379,087 records, both individually correlating above 0.75 with Total Score — the tree has little reason to split further on a competency that is one-third imputed median values, since imputed values carry no information for it to exploit. The near-zero importances for Place Value, Data Handling, and Fractions are an artefact of missingness concentrated in specific grades, not evidence that place value or data-handling ability doesn't matter.

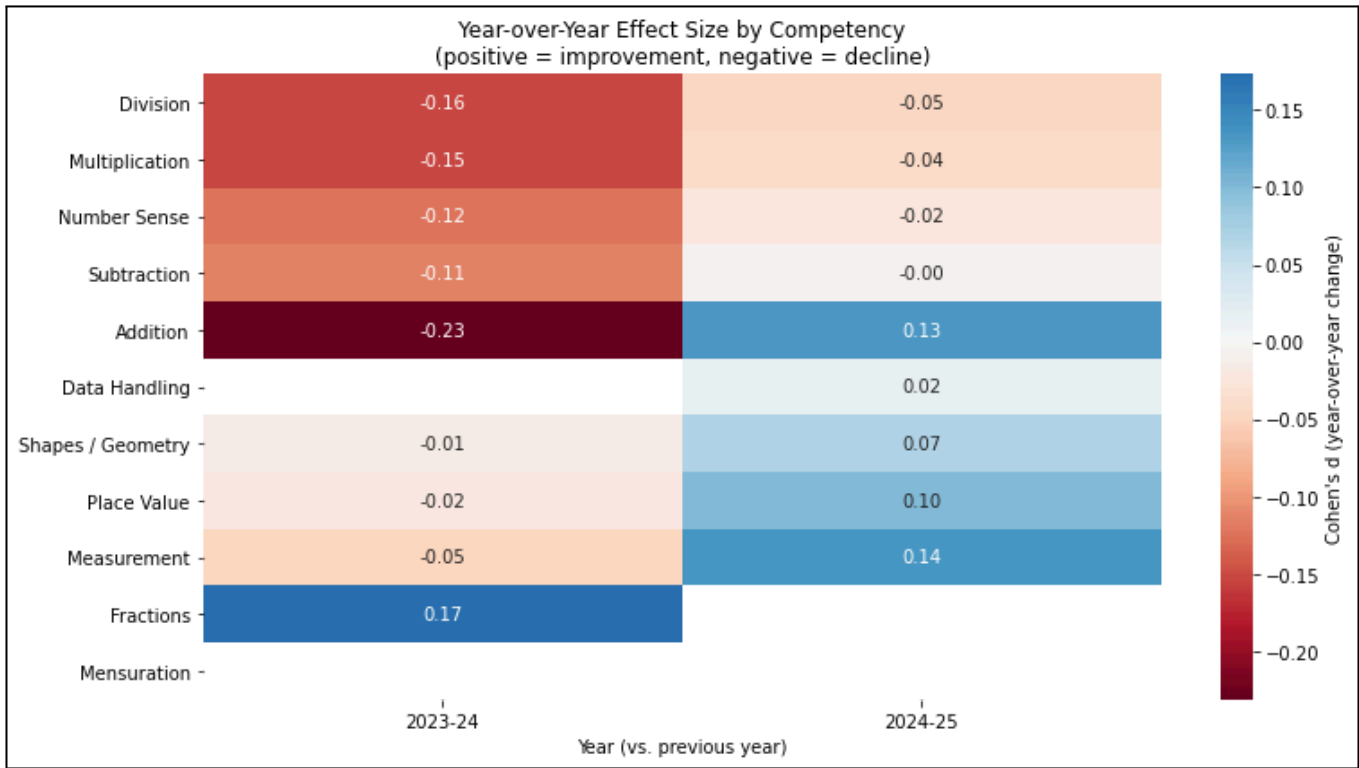
The practical reading across all three subsections: Division and Measurement are the two competencies that carry the most information about total performance, on every method that can see them cleanly, and they are also — from section 4.6 — where the test's internal correlation structure concentrates. The competencies with the weakest apparent effect (Place Value, Fractions, Mensuration) are the ones this dataset can least confidently speak to, because they are the ones least completely assessed, not necessarily the ones that matter least.

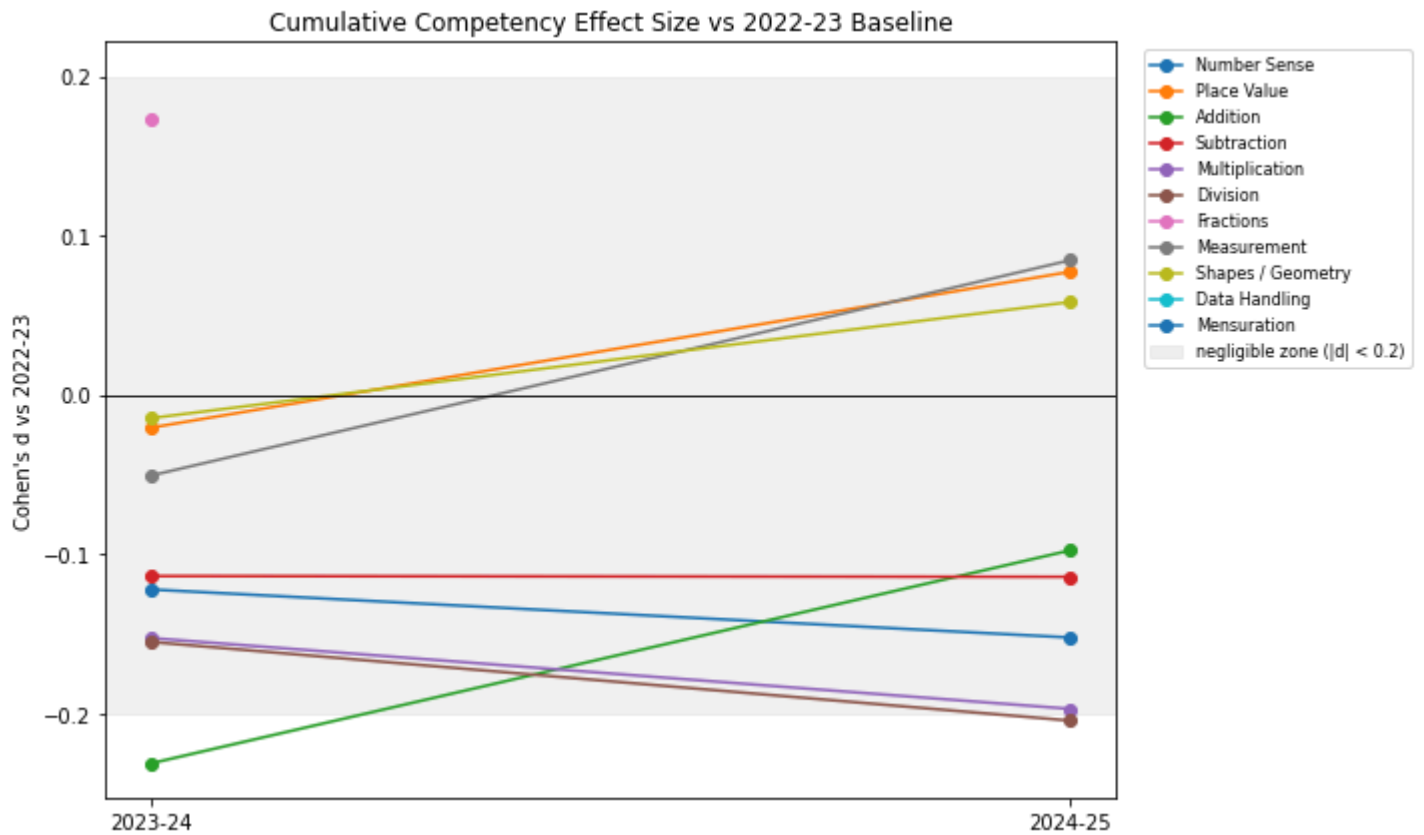
Year-over-year effect size, competency by competency

The Cohen's d view adds a dimension the raw percentage-point figures in section 2 don't show on their own: whether each competency's decline reversed, persisted, or was a one-off dip. Two consecutive year-pairs, computed on the full pooled dataset across all three grades, let each competency be classified as a genuine slide, a genuine recovery, or a dip that bounced back — three different stories that a single two-year comparison would collapse into one number.

Competency	2022-23 → 2023-24		2023-24 → 2024-25		Pattern
	Cohen's d	Direction	Cohen's d	Direction	
Addition	-0.231 (small)	declined	+0.133	improved	dip, then recovered
Division	-0.155	declined	-0.045	declined	persistent decline, both periods
Multiplication	-0.153	declined	-0.044	declined	persistent decline, both periods
Number Sense	-0.122	declined	-0.024	declined	persistent decline, both periods
Subtraction	-0.114	declined	-0.004	declined (flat)	decline, then plateau
Measurement	-0.051	declined	+0.135	improved	dip, then recovered
Place Value	-0.021	declined	+0.100	improved	dip, then recovered
Shapes / Geometry	-0.015	declined	+0.067	improved	dip, then recovered
Fractions	+0.173	improved	n/a (absent)	—	improved, then vanished from the instrument
Data Handling	n/a (absent in Y1)	—	+0.018	improved	introduced 2023-24, essentially flat since
Mensuration	n/a (absent)	—	n/a (single-year only)	—	exists only in 2024-25, no trend computable

Two competencies split cleanly from the rest: Division and Multiplication are the only ones that decline in *both* year-pairs rather than dipping and recovering. Number Sense follows the same pattern at a smaller magnitude. Everything else that fell in the first period — Addition, Measurement, Place Value, Shapes/Geometry — reversed in the second, several of them ending 2024-25 higher than they started. Subtraction sits in between: it stopped declining rather than recovering, essentially flat at -0.004 in the second period. This is a materially different picture from "scores fell, then partly came back everywhere." Most of the framework recovered. The two multiplicative competencies did not, and they are the same two competencies section 2.1 already identified as the site of the Grade 6 collapse and section 4.6 already identified as carrying the strongest signal in the total-score correlation structure. Three independent methods now point at the same pair.





The blanks are structural, not missing data in the ordinary sense, and match the instrument-coverage notes already established in section 2.3: Data Handling has no 2022-23 value because it wasn't part of that year's item set; Fractions has no 2023-24→2024-25 comparison because it was dropped from every 2024-25 file; Mensuration has no comparison at all because it exists only as a single grade-6 question introduced in 2024-25. A NaN in this table means the competency wasn't assessed in one of the two years being compared, not that no children attempted it.

Two caveats belong alongside this table, both in the direction of understatement rather than overstatement. First, every effect size here is "negligible" or, at most, "small" by conventional thresholds, including Addition's largest single-period value of -0.231. That reads as reassuring next to the double-digit percentage-point drops reported elsewhere in this document, but the two are not in tension: this table pools all three grades together, while the collapse is concentrated in Grade 6. Pooling a genuine, large decline in one grade against comparatively stable Grade 4 performance produces a diluted, pooled effect size — the same masking pattern the report has already guarded against in section 3 by testing the decline inside a fixed GP panel rather than trusting the raw pooled trend. Second, this table is computed on the raw, un-paneled sample, the same sample EH2 already showed can manufacture an apparent recovery (Grade 4 division's +5.6pp raw gain that becomes -0.6pp inside the panel). The Addition, Measurement, and Place Value recoveries shown here have not yet been run through that same panel check, and should be before they're reported as genuine improvement rather than a composition effect of who was assessed each year.

Recommendations

1. Launch a targeted Grade 5–6 multiplicative-reasoning remediation module (multiplication & division) in the lowest-performing GPs identified by the panel analysis, rather than a state-wide rollout.
2. Fix assessment coverage in the lowest-participation districts before drawing further conclusions from year-over-year trends; prioritize districts furthest from the 71% ceiling and any Kalyana Karnataka district lagging its regional peers.
3. Shift monitoring and resource-allocation dashboards from district-level to GP-level granularity, since district targeting reaches only a small fraction of total score variation.

Limitations

No school identifier, by design, so within-GP variation stays unattributed. Covariate vintages run 2011 to 2026 and are stated per table. Items changed across years within a constant competency framework, so cross-year statements are made at competency or ranking level, never as raw item-score gains, and the headline decline is corroborated by two external assessments. Coverage below 100% means tested children may differ from enrolled children. The competency mapping is the official per-file mapping supplied by the organisers. Duplicate rows are expected, since no persistent student ID exists, and are kept. GP and cluster do not nest, with 1,718 GP IDs spanning more than one cluster, so cluster is never treated as a child of GP. District aggregates for SC/ST and work columns use the Census CD-block universe, flagged in the data. All causal language is stated as consistent-with; this is observational data. Because of the absence of school level identification within GP clustering may further appear at school level, classroom level etc. As data was not available at these level we were not able to those relationships.

Sources

1. Akshara Foundation GP Maths Contest assessment (primary, on-site)
2. UDISE+ 2022-25 school and enrolment records
3. Census 2011 PCA via data.gov.in API
4. NFHS-5 (2019-21) and NFHS-6 (2023-24) fact sheets, IIPS and MoHFW
5. ASER 2024, ASER Centre
6. PGI-D 2025-26, Ministry of Education
7. PARAKH Rashtriya Sarvekshan 2024

Annexure

1. Recommendations table

#	Recommendation	Owner	Target Date	Est. Cost	Success Measure
---	----------------	-------	-------------	-----------	-----------------

1	Launch a targeted Grade 5–6 multiplicative-reasoning remediation module (multiplication & division) in the lowest-performing GPs identified by the panel analysis, rather than a state-wide rollout.	State/District Math Resource Persons, with Block Education Officers	Before start of next academic term	Low–moderate (existing resource-person cadre; module + printed materials, no new hires)	Grade 6 multiplication and division mean scores recover by a defined margin in the next assessment cycle, measured on the same fixed GP panel used here.
2	Fix assessment coverage in the lowest-participation districts before drawing further conclusions from year-over-year trends; prioritize districts furthest from the 71% ceiling and any Kalyana Karnataka district lagging its regional peers.	District Education Officers / District Assessment Coordinators	Within two assessment cycles	Low (operational/logistics — no new instrument needed)	State-wide participation rate rises meaningfully above the current 37.8% baseline, with no district below half the state average.
3	Shift monitoring and resource-allocation dashboards from district-level to GP-level granularity, since district targeting reaches only a small fraction of total score variation.	State Project Office / MIS & Data Team, with Cluster Resource Persons as field-level owners	Within one planning cycle (next budget/allocation round)	Low (reporting/dashboard change on data already collected; no new data collection)	A published GP-level performance dashboard is in active use for one full allocation cycle, and the share of variance addressed by targeting decisions increases from the district-only baseline.

